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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
19CS62C – PRINCIPLES OF COMPILER DESIGN [EVEN SEM 2024 – 2025]
LABORATORY EXERCISE – INSTRUCTION SHEET

Exercise 2: Create a symbol table, while recognizing identifiers

Problem Description:

A symbol table is a data structure used to store and manage information about symbols in a programming language, including their names, data types, scope, and memory locations. It is commonly used by compilers and interpreters to keep track of identifiers and their associated attributes and to perform tasks such as type checking, name resolution, and code generation.

It is utilized in the compiler's different phases, as shown below:

- **Lexical Analysis:** New table entries are created in the table, For example, entries about tokens.
- **Syntax Analysis:** Adds the information about attribute type,, dimension, scope line of reference, use, etc in the table.
- **Semantic Analysis:** Checks for semantics in the table, i.e., verifies that expressions and assignments are semantically accurate (type checking) and updates the table appropriately.
- **Intermediate Code generation:** The symbol table is used to determine how much and what type of run-time is allocated, as well as to add temporary variable data.
- **Code Optimization:** Uses information from the symbol table for machine-dependent optimization.
- **Target Code generation:** Uses the address information of the identifier in the table to generate code.

Each item in the symbol table is linked to a set of properties that assist the compiler at various stages.

Items stored in Symbol table:

- Labels in source languages
- Procedure and function names
- Variable names and constants
- Compiler generated temporaries
- Literal constants and strings

Information used by the compiler from the Symbol table:

- Data type and name
- Offset in storage
- Declaring procedures
- For parameters, whether parameter passing by value or by reference
- Number and type of arguments passed to function
- Base Address

Your task is to build a symbol table to track the variable name and its associate properties in the source program for the following situation:

- **After scanning of declaration statement**
- **After processing input statement where the variables value updated**
- **After processing calculation / assignment of value to variable**

Sample Input File:

```
#include <stdio.h>
void main()
{
int a,b;
printf("Enter value for a:");
scanf("%d",&a);
/* calculation of b */
b=a*5;
/*swapping of values a and b */
a = a + b;
b = a – b;
a = a – b;
}
```

Output:

After processing declaration statement

Symbol Table Details are:

var-name	d-type	size	address	Prev-value	Curr-value
a	int	4	1419463688	0	0
b	int	4	1419463692	0	0

Enter value for a: 20

After processing input statement

Symbol Table Details are:

var-name	d-type	size	address	Prev-value	Curr-value
a	int	4	1419463688	0	20
b	int	4	1419463692	0	0

After processing calculation statement

Symbol Table Details are:

var-name	d-type	size	address	Prev-value	Curr-value
a	int	4	1419463688	0	20
b	int	4	1419463692	0	100

After swapping of values statement

Symbol Table Details are:

var-name	d-type	size	address	Prev-value	Curr-value
a	int	4	1419463688	20	100
b	int	4	1419463692	100	20

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